

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your latest score.

Next item →

Instructions

1.

Neural network is a computational graph.

True

False

✔

Correct

A neural network is represented as a computational graph with nodes for operations and edges for data flow.
- 1 / 1 point

2.

Shuffling the input data helps the greedy optimization.

False

True

✔

Correct

Shuffling input data prevents bias and improves the generalization of greedy optimization.

1 / 1 point

3.

The major optimizer in NN is the gradient descent.

True

False

✔

Correct

Gradient descent is the primary optimizer used in neural networks to minimize the loss function by adjusting weights.

1 / 1 point

4.

The perceptron model can solve the XOR problem via kernels.

True

False

✘

Incorrect

Kernels are used by SVM.

1 point

5.

Neural networks are greedy optimizers.

False

True

✔

Correct

Neural networks use gradient-based methods that iteratively make local improvements to minimize loss.

1 / 1 point

6.

A deep neural network has more than 1 hidden layers.

False

True

✔

Correct

A deep neural network has more than one hidden layer, distinguishing it from shallow networks with a single hidden layer.

1 / 1 point

7.

What is the purpose of the activation function in a neural network?

To update weights

To introduce non-linearity into the model

To compute loss

To perform matrix multiplication

✔

Correct

Activation functions enable neural networks to learn complex patterns by introducing non-linear transformations.

1 / 1 point

8.

What is the term used to describe the ability of a neural network to generalize well to new, unseen data?

Regularization

Overfitting

Generalization

Convergence

✔

Correct

Generalization refers to the neural network's ability to perform well on new, unseen data.

1 / 1 point

9.

Neural network is an excellent classifier when the number of data points << the number of features.

False

True

✔

Correct

NN is worst and SVM is best in that case.

1 / 1 point

10.

Neural networks back-propagation algorithm carries the error from the input to the output layer.

True

False

✔

Correct

Error is carried from output to input.

1 / 1 point